

Insights Into Intra-Domain SAV Architecture/Solution Design

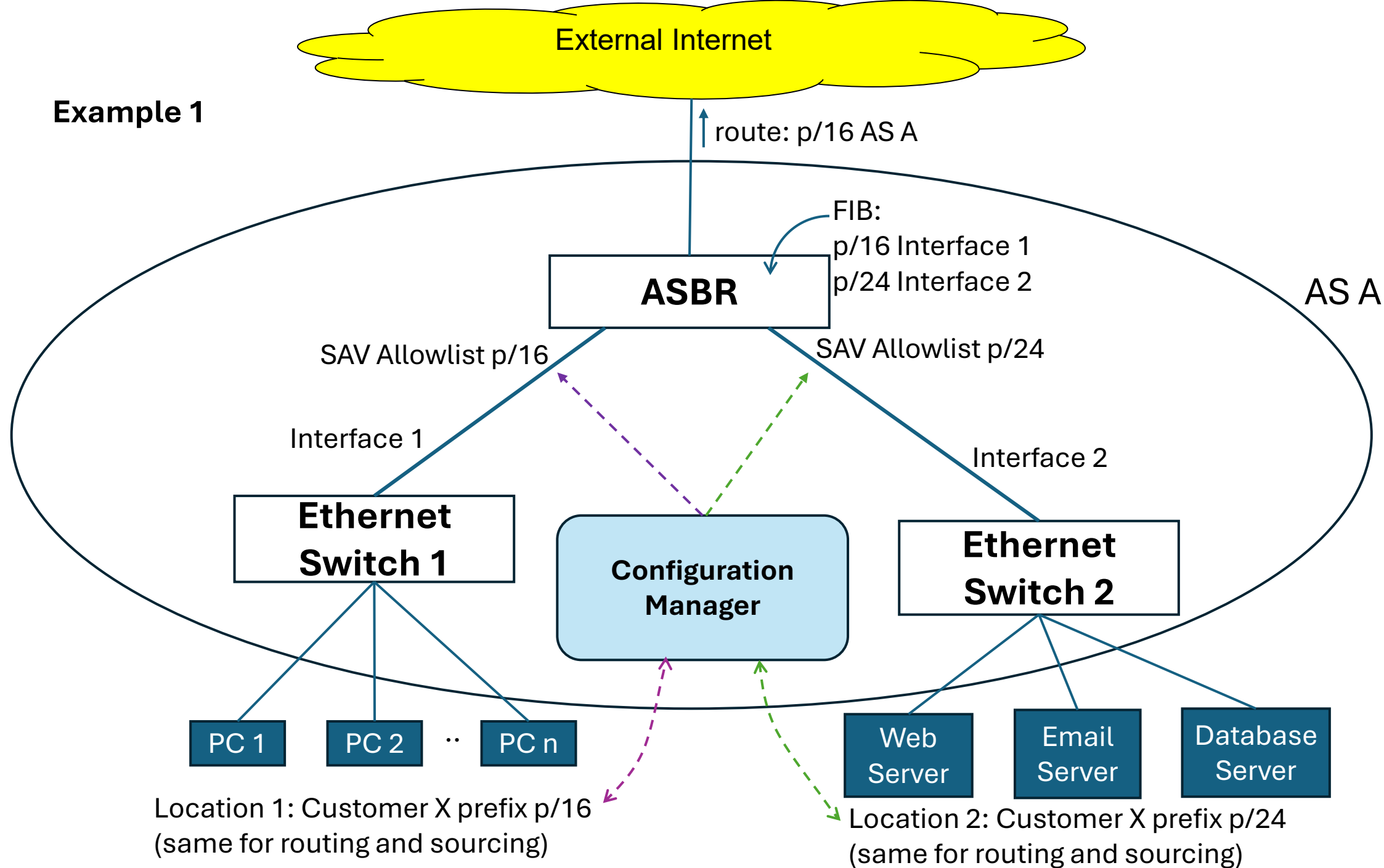
K. Sriram

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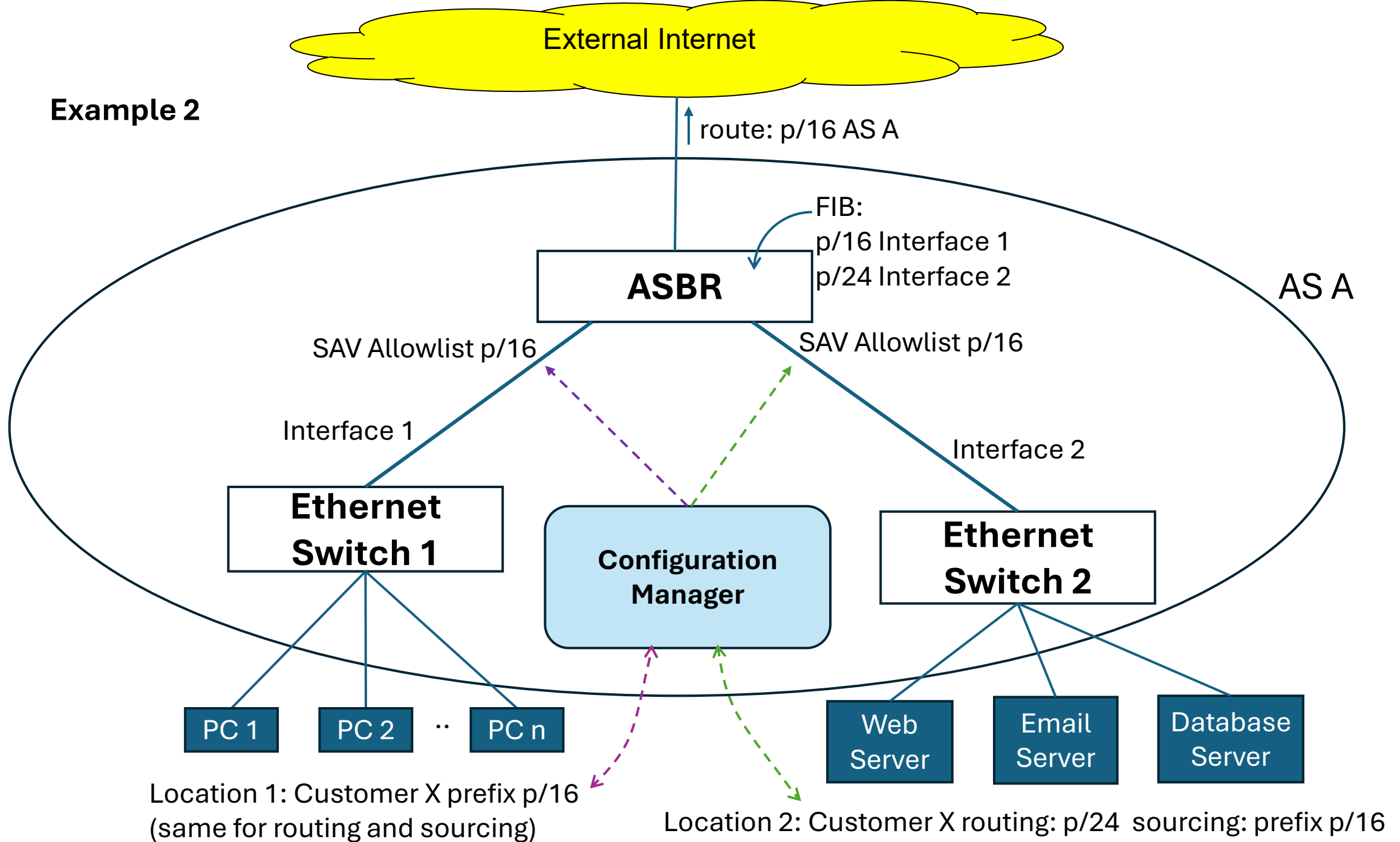
Key Top-Level Insights

- Configuration information is sufficient (for intra-domain SAV)
 - exchanged between the AS and the customer (non-AS)
- Customer does not generate/send any routing messages (e.g., BGP Updates)
- Any routing information exchanged between ASBRs/routers follows from the configuration information
- These intra-AS routing messages and resulting FIB entries do not contain customer identification info
 - Hence, cannot be used to modify SAV tables for customer interfaces
- If there is any asymmetry in routing/sourcing prefixes used by a customer, it is disclosed as part of the configuration information for each interface involved
- Hidden prefix issue does not arise – customer can directly disclose to the configuration manager any non-routed prefix used for sourcing
- DSR prefixes can also be directly disclosed to the configuration manager
- ROA information is also secondary to configuration information; generate an alert if the configuration information does not match the ROA

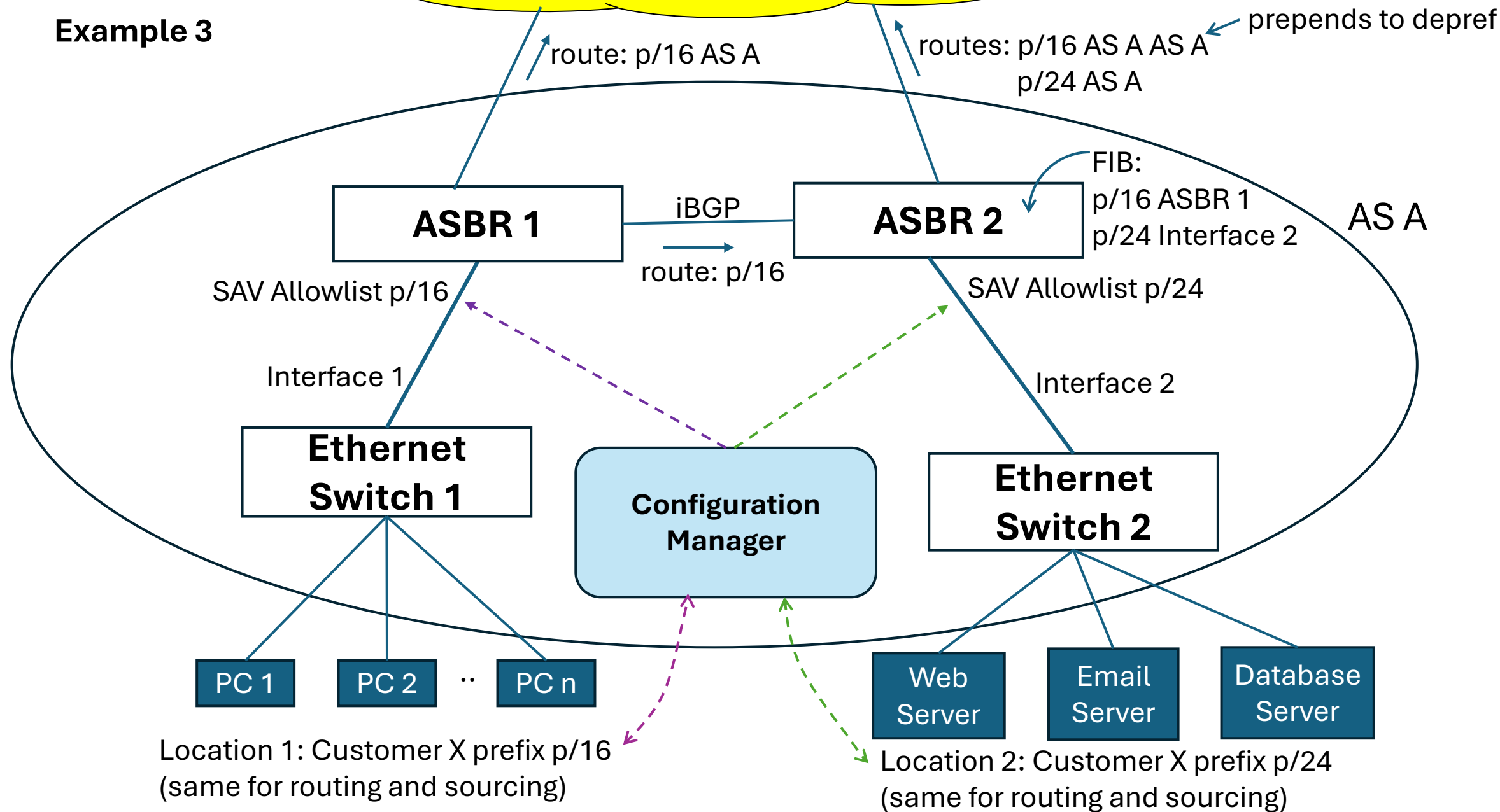
Example 1



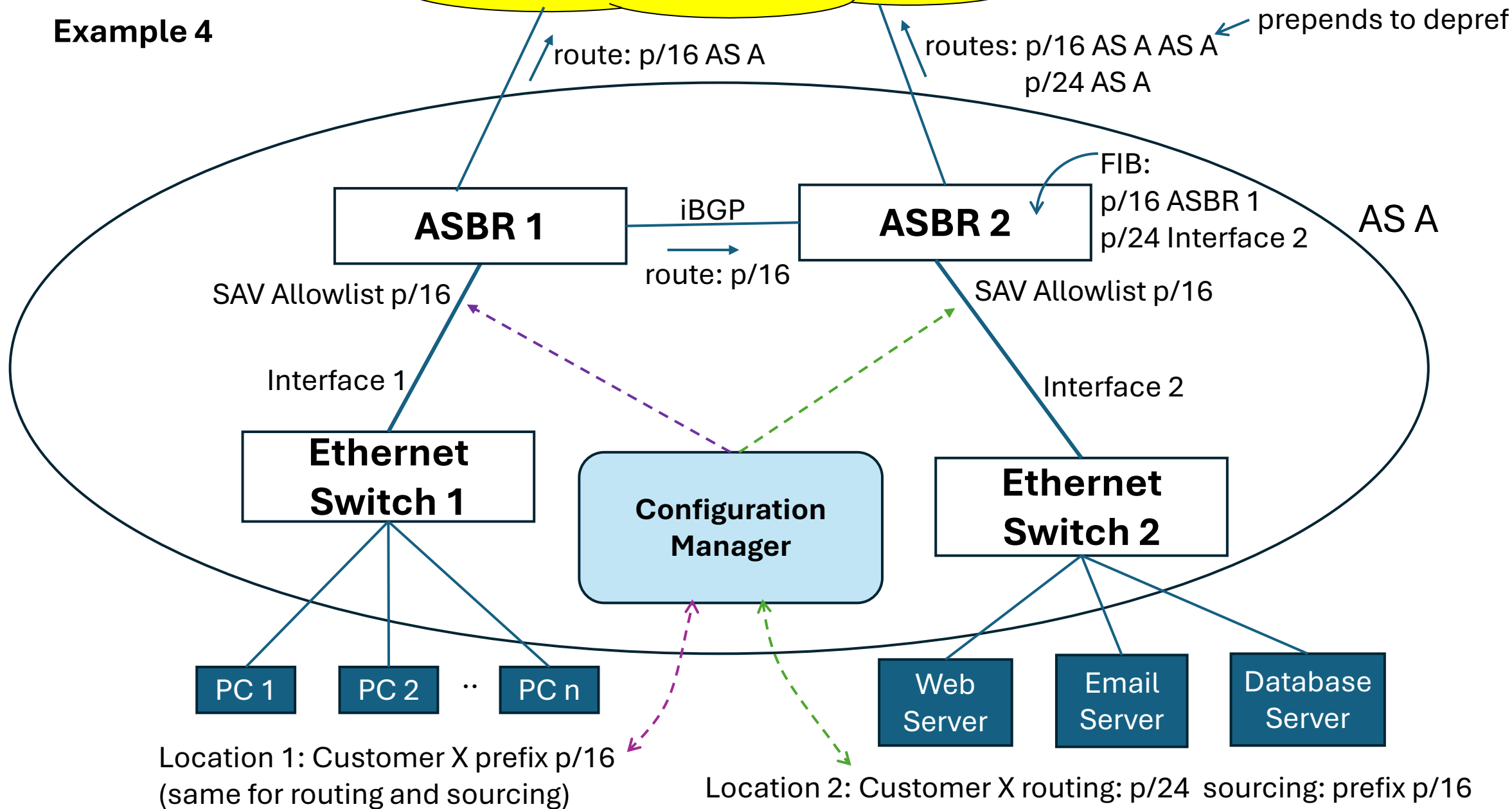
Example 2



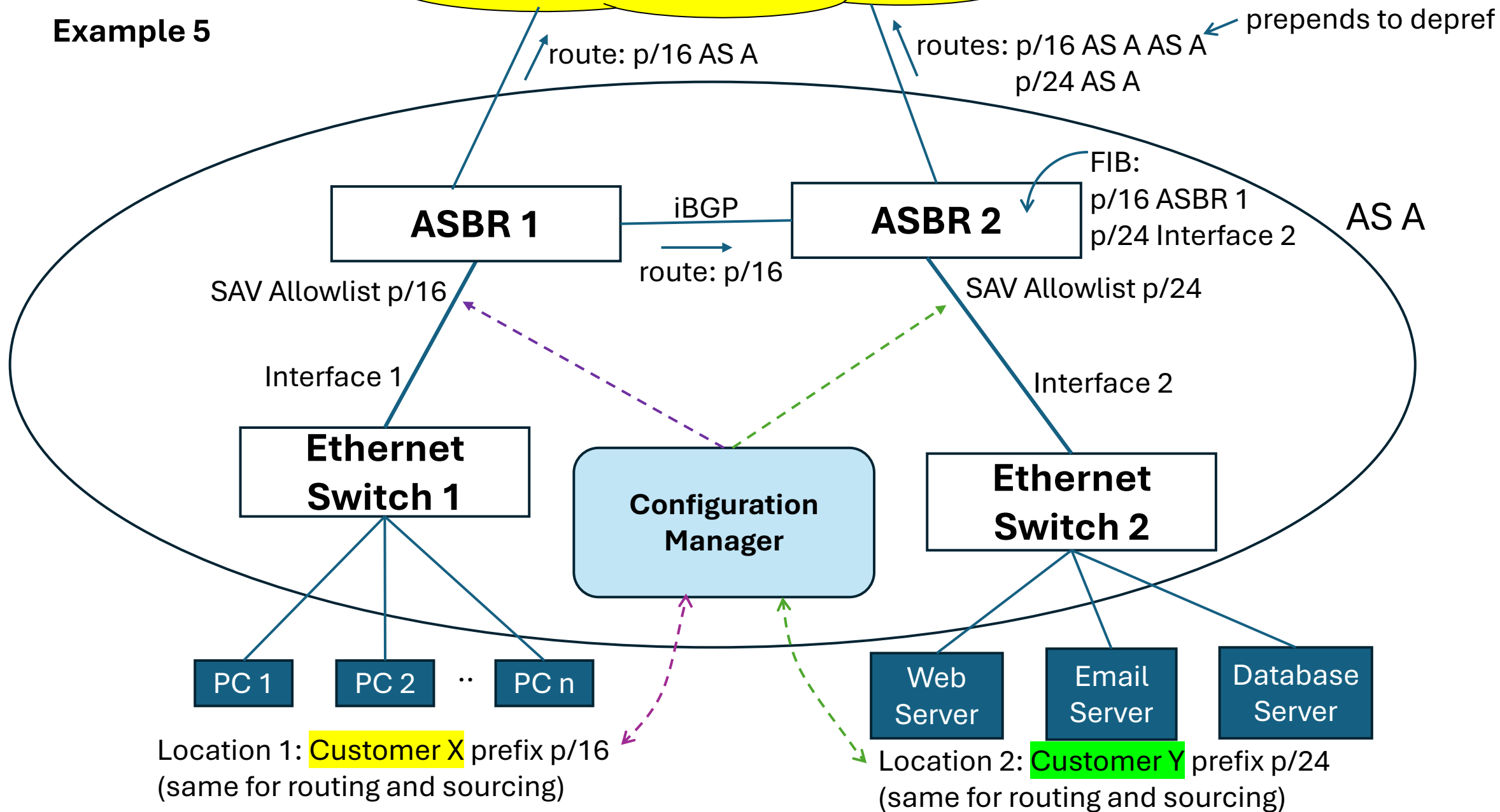
Example 3



Example 4



Example 5



Key Points of Observation from Examples 1 – 5

- There are no routing messages (like BGP Updates) from the customers in intra-domain (i.e., originating-AS or leaf AS) scenarios
- So, routes in the FIBs corresponding to interfaces facing customers are necessarily configured based on info from the configuration manager
- Customers must inform the configuration manager what prefixes they want announced in routing and what prefixes they intend to use for sourcing on a given interface where they are connected
 - Details concerning asymmetry in routing/sourcing should be explicitly conveyed by customers during the configuration
 - If they do not separately specify source prefixes, then route prefixes should be assumed to be source prefixes as well – symmetry assumption
- Key point: customer (“entity”) routes in the FIB that came via another ASBR (or router) DO NOT have any marking that identifies the customer (“entity”)
 - So, such routing information cannot be used to modify SAV tables (this indirection can produce errors in the SAV tables – see Example 5 in slide 8)
- Thus, SAV tables (allowlists) for customer interfaces must be derived from the configuration information alone for intra-domain SAV

There is No Hidden Prefixes Issue in the Intra- Scenarios

- Hidden prefix issue may arise in inter-domain SAV due to NO_EXPORT but not in intra-domain SAV
 - Customers disclose their prefixes (intended to be used for routing or sourcing) as part of the configuration procedure with the provider AS (i.e., originating AS)
 - If there is asymmetry, it is even more important that they disclose the prefixes that are not routed but used for sourcing at a given interface with the AS
- **DSR scenario** is easily handled in intra-domain SAV through configuration
 - Customers disclose such a prefix during configuration as used for sourcing but not routed
 - ROA should be registered for the prefix, but ROA does not have precedence over the configuration information because some unrelated prefix owner may have created a ROA erroneously (sometimes malicious) showing the AS in consideration as the origin AS. The AS in consideration can detect and report such ROAs because it has reliable local configuration information that it can check against.

ROA is Used for Cross-Checking Configuration Info

